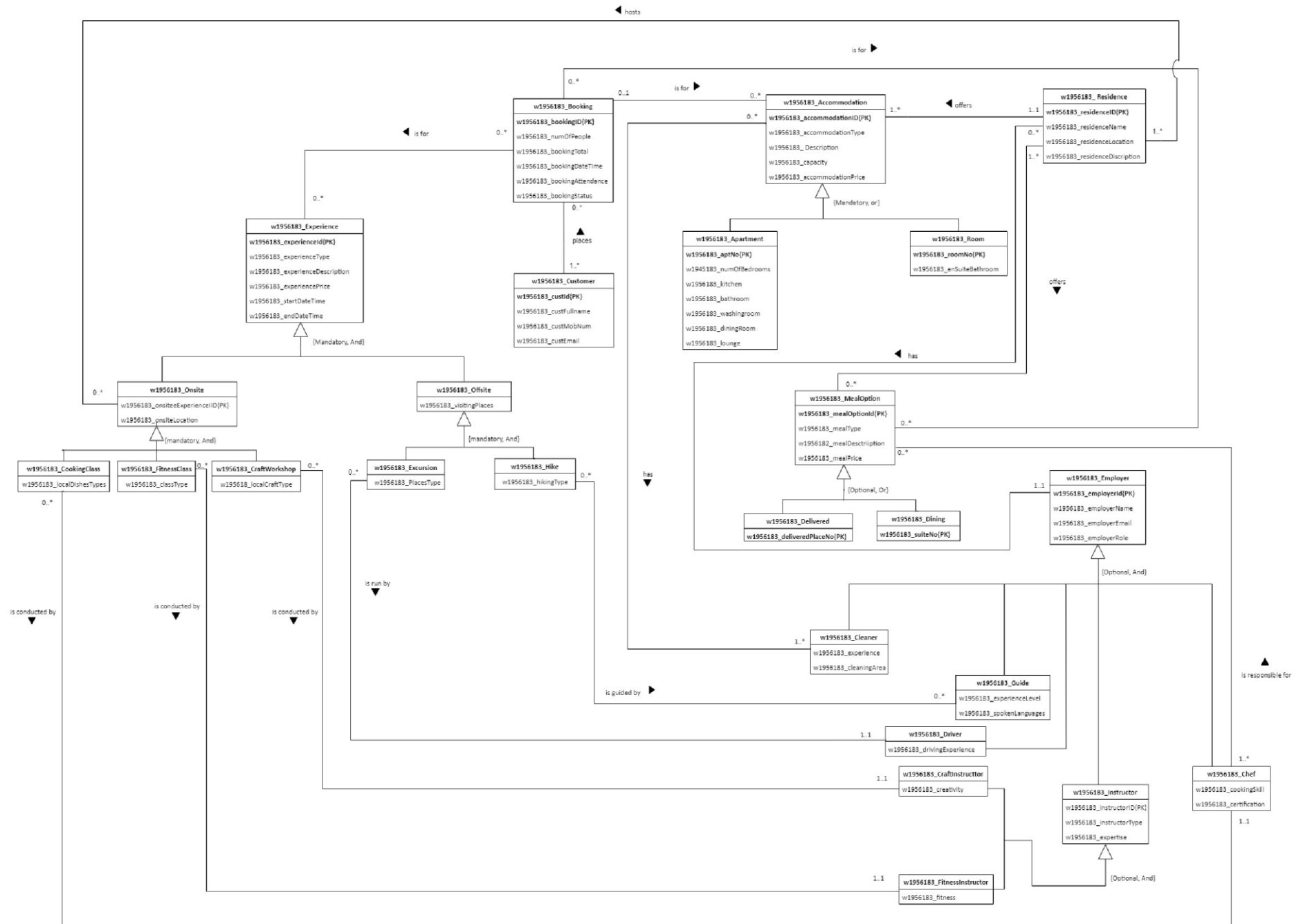

Database Systems Design & UML Modeling for Tourism Management

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Project Overview

This project focuses on the design and modeling of conceptual and logical database systems using UML notations for tourism management scenarios. The work involved identifying entities, relationships, multiplicities, attributes, primary and foreign keys, and transforming business requirements into structured ERD models. The project demonstrates database design principles, system analysis, relational modeling, and logical data organization using industry-standard UML methodologies.

EERD Diagram



Data Dictionary tables

Entity Data Dictionary Table

Entity name	Brief Explanation
Customer	Represents individuals booking holidays with Vacoliday.
Booking	Represents the reservation made by a customer for accommodations, meals, and experiences
Residence	Represents the accommodations provided by Vacoliday, including details about locations and types.
Accommodation	Represents bookable units within residences, with subclasses: Fully-Integrated Apartment, Stand-Alone Room.
MealOption	Represents various meal options offered by Vacoliday, with subclasses: Delivered, Dining
Experience	Represents bookable experiences, with subclasses: On-Site Experience, OffSite Experience.
OnSiteExperience	Represents on-site experiences offered within a specific residence, with subclasses: Fitness Class, Craft Workshop, Cooking Class.
OffSiteExperience	Represents off-site experiences led by a guide, with subclasses: Excursion, Walking Hike.
Employer	Represents the staff members employed by Vacoliday, with subclasses: Guide, Driver, Chef, Cleaner, Instructor.
Instructor	Represents instructors with subclasses leading fitness instructor and craft workshop instructor

Specialisation Data Dictionary Table

General Entity	Specialised Entity	Brief Explanation
Accommodation	Fully-Integrated Apartment	Represents apartments with multiple rooms, kitchens, and lounges.
Accommodation	Stand-Alone Room	Represents single rooms, possibly with an en-suite bathroom.
Experience	On-Site Experience	Represents experiences taking place within a specific residence, including Fitness Class, Craft Workshop, and Cooking Class.
On-Site Experience	Fitness Class	Represents on-site fitness classes guided by a fitness instructor.
On-Site Experience	Craft Workshop	Represents on-site craft workshops with a workshop instructor.
On-Site Experience	Cooking Class	Represents on-site cooking classes taught by a chef.
Experience	Off-Site Experience	Represents experiences conducted outside the residence, including Excursion and Walking Hike.
Employee	Guide	Represents employees overseeing off-site experiences.
Employee	Driver	Represents employees using buses for excursions.
Employee	Chef	Represents employees responsible for meals and cooking classes.
Employee	Housekeeping Attendant	Represents employees responsible for the maintenance of accommodations.
Employee	Instructor	Represents employees leading fitness and craft workshops.
MealOption	Dining	Represents meal options that are taken collectively at dining suites within the Vacoliday residences.
MealOption	Delivered	Represents meal options that can be delivered to people's accommodations if pre-booked in advance.
Instructor	Craft Instructor	Represents employees who specifically lead craft workshops.

Instructor	Fitness Instructor	Represents employees who specifically lead fitness classes.
Off-Site Experience	Excursion	Represents off-site excursion drive by driver
Off-Site Experience	Hike	Represents off-site hike guided by guide

ERD Diagram

